

Exact Mixed-Norm Bernstein Numbers in the Aligned Parameter Regions

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Source Question

The source is Chuah–Lang–Yao, *A Classification Theorem on Non-compact Embeddings between Besov Spaces*, arXiv:2410.10731 [1]. For

$$0 < p_0 < p_1 \leq \infty, \quad 0 < q_0 < q_1 \leq \infty,$$

Proposition 3.5 proves that the formal embedding

$$I : \ell^{q_0}(\ell^{p_0}) \longrightarrow \ell^{q_1}(\ell^{p_1})$$

satisfies

$$b_n(I) \leq n^{-\min(1/p_0, 1/q_0)(1-\max(p_0/p_1, q_0/q_1))}. \quad (1)$$

Section 6 asks for sharper upper bounds and matching lower bounds, and singles out the decay of these mixed-sequence Bernstein numbers as the critical question for noncompact Besov embeddings on $\mathbb{R}^d$. More precisely, the source writes:

“We believe that the upper bounds in Proposition 3.5 could be sharpened, and that it should be possible to derive corresponding lower bounds.” It then identifies the case $\Omega = \mathbb{R}^d$ with computing the decay of $b_n(\ell^{q_0}(\ell^{p_0}) \rightarrow \ell^{q_1}(\ell^{p_1}))$.

6. FURTHER REMARKS

We conclude our paper with a few remarks and open questions.

In Theorems 1.1 and 1.2, we observe the impact of differences between function spaces on bounded Lipschitz domains Ω and on $\mathbb{R}^n$. A natural question that arises is what results can be obtained for embeddings on unbounded open sets (say unbounded uniform domains).

We believe that the upper bounds in Proposition 3.5 could be sharpened, and that it should be possible to derive corresponding lower bounds. This could lead to improved estimates on the rate of decay for Bernstein numbers between Besov spaces when the embedding is non-compact but finitely strictly singular. Note that although we do not have a canonical definition of Besov norms, the decay rates of their Bernstein numbers are determined up to a constant multiple which still make sense. The case where $\Omega = \mathbb{R}^n$ is particularly intriguing, which is equivalent to computing the decay rate of $b_n(\ell^{q_0}(\ell^{p_0}) \rightarrow \ell^{q_1}(\ell^{p_1}))$.

Here $1/\infty = 0$ and $r/\infty = 0$. We use the source convention

$$b_n(T) = \sup_{\dim E=n} \inf_{0 \neq x \in E} \frac{\|Tx\|}{\|x\|}.$$

Proof Intuition

There are two canonical n -dimensional tests. Matrices supported in one row see only the inner p -embedding; matrices supported in one column see only the outer q -embedding. They give a lower bound governed by the slower of the two scalar ℓ^r decay rates. The source upper exponent is the product of two minima. When the smaller initial reciprocal exponent and the smaller relative loss occur in the same coordinate direction, that product is exactly the slower scalar rate, so the upper and lower bounds coincide without any loss. In crossed regions the two minima come from different directions; this argument no longer closes the gap, and those regions remain open in this packet.

Result

Theorem 1. *Suppose, in addition, that either*

$$p_0 \geq q_0 \quad \text{and} \quad \frac{p_0}{p_1} \geq \frac{q_0}{q_1}, \quad (2)$$

or

$$p_0 \leq q_0 \quad \text{and} \quad \frac{p_0}{p_1} \leq \frac{q_0}{q_1}. \quad (3)$$

Then, for every $n \geq 1$,

$$b_n(I) = n^{-\min\{1/p_0-1/p_1, 1/q_0-1/q_1\}}. \quad (4)$$

Thus the source upper bound is already sharp, with constant one, throughout the two aligned parameter regions (2)–(3).

The alignment says that the smaller of $1/p_0, 1/q_0$ and the smaller of the two relative exponent losses come from the same coordinate direction. It includes, for example, all cases $p_0 = q_0$ and all cases $p_0/p_1 = q_0/q_1$.

Proof

Put

$$a = \frac{1}{p_0}, \quad c = \frac{1}{q_0}, \quad u = 1 - \frac{p_0}{p_1}, \quad v = 1 - \frac{q_0}{q_1}.$$

The two one-direction coordinate subspaces give the complete lower-bound argument. Fix one outer index and let E_p consist of matrices supported on the first n inner coordinates of that row. For $x \in E_p$ the outer norms disappear, and the standard finite-dimensional norm comparison gives

$$\frac{\|x\|_{\ell^{q_1}(\ell^{p_1})}}{\|x\|_{\ell^{q_0}(\ell^{p_0})}} = \frac{\|x\|_{p_1}}{\|x\|_{p_0}} \geq n^{1/p_1-1/p_0} = n^{-au}.$$

Equality is attained by a constant-modulus vector, so the minimum modulus on E_p is exactly n^{-au} . Consequently $b_n(I) \geq n^{-au}$.

Next fix one inner coordinate and use the first n outer indices. The same argument with q_0, q_1 gives $b_n(I) \geq n^{-cv}$. Therefore

$$b_n(I) \geq \max\{n^{-au}, n^{-cv}\} = n^{-\min(au, cv)}. \quad (5)$$

The exponent in the source upper bound (1) is

$$\min(a, c)(1 - \max(1 - u, 1 - v)) = \min(a, c) \min(u, v). \quad (6)$$

Under (2), one has $a \leq c$ and $u \leq v$, hence

$$\min(a, c) \min(u, v) = au = \min(au, cv).$$

Under (3), one has $c \leq a$ and $v \leq u$, hence both sides equal cv . Thus (1) and (5) coincide, which proves (4). $\square$

Besov Consequence

Corollary 1. *Consider a critical noncompact embedding on $\mathbb{R}^d$ covered by the source wavelet reduction, with $p_0 < p_1$, $q_0 < q_1$, and*

$$s_0 - \frac{d}{p_0} = s_1 - \frac{d}{p_1}.$$

If (2) or (3) holds, then its Bernstein numbers have the exact power-law order

$$b_n \asymp n^{-\min\{1/p_0-1/p_1, 1/q_0-1/q_1\}}.$$

The implicit constants come only from the wavelet isomorphisms defining the equivalent Besov norms.

Proof. The source identifies both Besov spaces with weighted mixed-sequence spaces. At the displayed critical relation the same diagonal weight occurs on both sides and is removed by a common isometry. The embedding is therefore conjugate, up to the fixed wavelet isomorphisms, to I in Theorem 1. Bernstein numbers are stable under composition with isomorphisms, giving the assertion. $\square$

Why the Crossed Regions Remain Open

If, say, $a < c$ but $u > v$, then

$$\min(a, c) \min(u, v) = av < \min(au, cv).$$

Thus the source upper estimate and the coordinate lower bound have a real gap. Six focused upgrade attempts were made: retaining the row-count profile in the repeated-maximum lemma; a row/column projection-rank dichotomy; transposition via mixed-norm Minkowski inequalities; factorization through intermediate mixed spaces; flattening through a single ℓ^r ; and rectangular/random-sign subspaces. None closes the gap.

The obstruction is dimension allocation. A repeated-maximum vector may hide near-maximal coordinates in an unrecorded row, while for an entire n -dimensional subspace that row has limited projection rank. A lossless rank/kernel recursion could plausibly prove the full coordinate exponent, but the elementary iteration loses dimension and decay simultaneously. Accordingly, this packet claims only the aligned theorem.

Verification and Novelty Status

The proof is an exact exponent comparison plus two coordinate restrictions; a rational-grid script checks the algebra independently. The source TeX, run indexes, exact-formula searches, and mixed-norm/Bernstein-number searches were checked. No later full solution or prior statement of this equality was found in that bounded search. The status is *candidate new substantial partial result*, not an exhaustive priority claim.

Human-Review Recommendation

Review the finite ℓ^p minimum modulus, the alignment equivalences, and the transfer through the source wavelet isomorphisms. No claim is made in either crossed region.

References

- [1] C. Y. Chuah, J. Lang, and L. Yao, *A Classification Theorem on Non-compact Embeddings between Besov Spaces*, arXiv:2410.10731.